



Disaster prevention learning materials

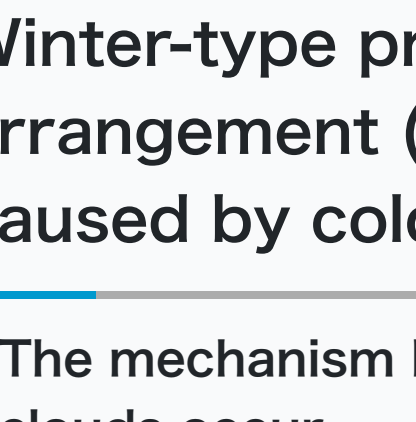
Characteristics of heavy snow in Ishikari, Sorachi, and Goshi regions

Things to be aware of during heavy snow

- ▶ When heavy snow warnings and advisories are issued, there is a risk of traffic obstructions due to heavy snow, isolation of settlements, human damage from cars trapped in snow, and disasters caused by avalanches on slopes.
In particular, when a storm warning or a wind and snow warning is issued, and accompanied by storms and strong winds, localized forecasts and blows will occur, and the risk of traffic obstructions will increase.
- ▶ When a snowfall warning is issued, please be careful of snowfall on power lines, communication lines, etc.

Snow caused by developed low pressure

- ▶ When the low pressure is in the west, winds from the south blow and wet heavy snow fall. At this time, you need to be careful of snow and traffic obstruction caused by wet snow.
- ▶ On the other hand, when the low pressure moves east, winds from the north blow and turn into dry snow. At this time, it is necessary to pay attention to the decline in outlook and blowing up due to fierce fubuki.



Characteristics of snow due to the position of low pressure and wind direction

Other features

- ▶ Around the low pressure, the slope of the pressure is large (the isobaric lines are dense), the wind is strong, and the waves are high in the sea.
- ▶ Due to the movement of low pressure, the wind direction and temperature change rapidly.

You also need to pay attention to the following things!

- ▶ Depending on the place, the wind may intense sharply.
- ▶ The accumulation of snow of different qualities makes it easier for avalanches to occur on slopes.

Winter-type pressure arrangement (sindial clouds caused by cold air)

The mechanism by which snow clouds occur

- ▶ The cold, dry air blowing from the Siberian continent flows into the Sea of Japan.
- ▶ Clouds are generated by receiving water vapor near the sea surface of the Sea of Japan.
- ▶ The cold air is strong, and the higher the seawater temperature in the Sea of Japan, the more snow clouds develop.
- ▶ The snow clouds make it snow on the Sea of Japan side.



The mechanism that causes snow to fall on the Sea of Japan side

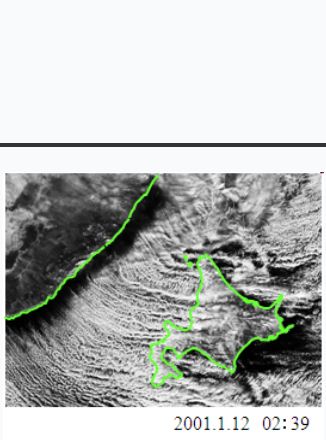
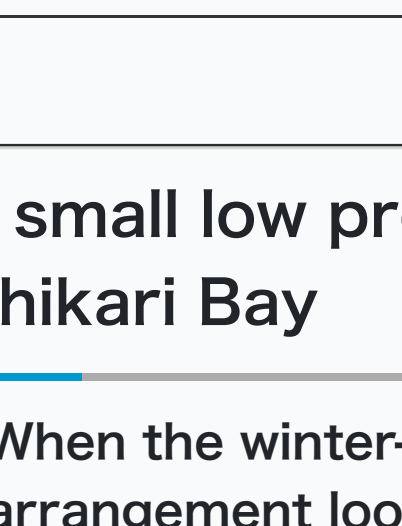
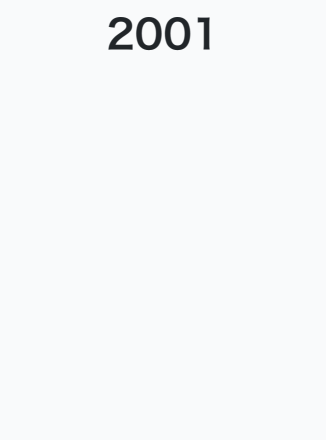
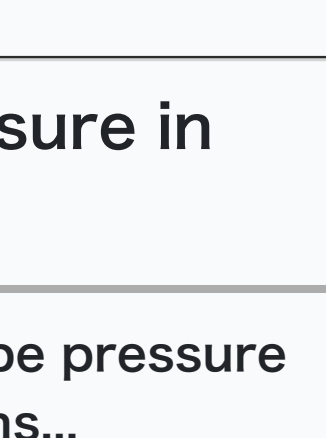
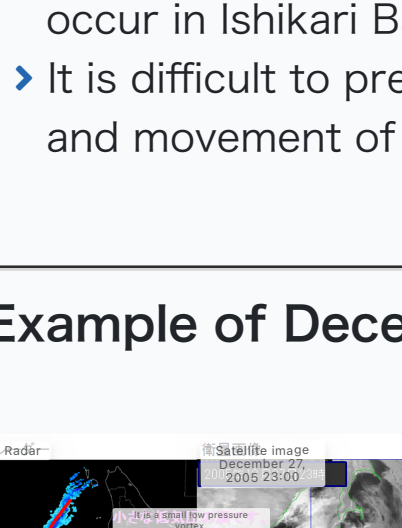
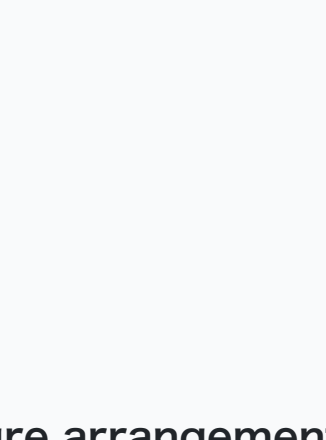
If there is a high pressure in the west and a low pressure in the east...



High pressure in the west of Hokkaido, low pressure in the east

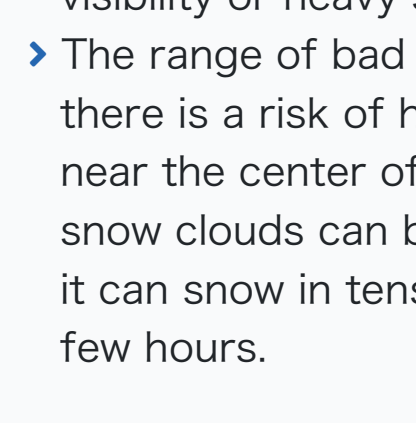
- ▶ The north-to-west wind from the Sea of Japan makes it easier for snow clouds on the sea to enter.
- ▶ Depending on the wind direction, the area that is easy to enter varies.
- ▶ In addition, when the wind speed is weak, a lot of snow falls in the coastal region, and as the wind speed increases, snow clouds tend to move inland.
- ▶ Due to the long-term heavy snow, it is necessary to pay attention to local heavy snow and poor prospects.

Wind direction and snowy areas

Areas with more wind direction and snow	Satellite image
In the case of north-northwest wind  Areas where snow is more due to the north-northwest wind Central and southern Ishikari Northern part of Goshi, foothills of Mt. Yotei And there will be more snow.	 Satellite image at 12:40 on February 1, 2009
In the case of the northwest wind  Areas where snow increases due to the northwest wind Ishikari region Kita-Sorachi and Minami-Sora-chi Northern part of Goshi, foothills of Mt. Yotei And there will be more snow.	 Satellite image at 02:39 on December 17, 1998
In the case of northwest west wind  Areas where there is more snow due to the northwest wind Kita-Sorachi and Minami-Sora-chi Northern Ishikari Foot of Mt. Yotei And there will be more snow. It is often sunny in the center of Sapporo, It is easy to snow in Jozankei.	 Satellite image at 02:39 on February 11, 2001
In the case of west wind  Areas where there is a lot of snow due to the west wind Sorachi region Northern Ishikari Foot of Mt. Yotei And there will be more snow.	 Satellite image at 02:39 on January 12, 2001

A small low pressure in Ishikari Bay

When the winter-type pressure arrangement loosens...



The winter-type pressure arrangement weakens, and a small low pressure occurs in Ishikari Bay

- ▶ When the winter pressure arrangement weakens, a small low pressure may occur in Ishikari Bay.
- ▶ It is difficult to predict the occurrence and movement of small low pressure.

Example of December 27, 2005

A small low pressure vortex occurs near Ishikari Bay

The main thing to pay attention to

- ▶ In the south to southwest of a small low pressure system, sudden winds and snow may lead to a decrease in visibility or heavy snowfall.
- ▶ The range of bad weather is local, but there is a risk of heavy snowstorms near the center of the vortex, and snow clouds can be concentrated, and it can snow in tens of centimeters in a few hours.